Assessing the Financial and Coverage Effects of Medicaid Expansion

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Group 3

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Section One: Introduction

Access to affordable health insurance is important for economic stability and public health. Yet prior to the Affordable Care Act (ACA), millions of low-income adults faced exclusion from insurance markets due to unaffordable premiums, pre-existing condition clauses, and restrictive Medicaid eligibility rules. The ACA's Medicaid expansion (2014)—which extended coverage to adults earning ≤138% of the federal poverty level (FPL)—created a natural experiment to study how policy interventions transform insurance access, affordability, and health behaviors. This project evaluates the causal effects of Medicaid expansion on economic outcomes: insurance coverage, premium costs, and OOP burdens. Leveraging quasi-experimental variation between expansion and non-expansion states, we employ difference-in-differences (DiD) method and seemingly unrelated regression (SUR) joint significance test to study the policy's impacts. Our analysis builds on Grossman's (1972) health capital model, which posits that insurance lowers the marginal cost of health investments (π) and addresses Arrow's (1963) idea on market failures in healthcare. For policymakers, our findings inform debates about the ACA's efficacy, the trade-offs between public and private coverage, and strategies to strengthen safety nets. By quantifying how Medicaid expansion reshaped financial protection and access, we contribute evidence to guide future reforms—whether scaling expansion in holdout states or designing complementary programs to expand health gains.

Section Two: Literature review

Recent research has consistently demonstrated the success of Medicaid expansion in increasing insurance coverage. Seminal work by Sommers et al. (2014) demonstrated significant reductions in uninsured rates, with particularly strong impacts in expansion states, and Courtemanche et al. (2017) refined these findings through quasi-experimental designs, showing that expansion states achieved nearly double the coverage gains of non-expansion states. Subsequent studies by Courtemanche et al. (2019, 2020) further established that these improvements reduced socioeconomic disparities, with effects persisting through 2018.

The literature has also examined behavioral and financial responses to expanded coverage. The Oregon Health Insurance Experiment (Finkelstein et al., 2012) found that Medicaid significantly reduced out-of-pocket spending. Simon et al. (2016) adopted SUR for testing joint significance of Medicaid expansion effects across multiple correlated health outcomes within a DiD framework. They used conventional DiD regressions for individual outcome estimates but employed SUR to assess whether the expansion simultaneously affected insurance coverage, preventive care utilization, and health behaviors. Their joint significance tests showed that expanded coverage increased preventive care without inducing moral hazard - a finding that contrasted with Courtemanche et al. (2018), who reported simultaneous rises in both preventive service use and risky behaviors.

Although existing studies have examined the effects of insurance expansions on financial outcomes, they analyzed these outcomes in isolation and overlooked potential interdependencies. To address this gap, we follow a methodology similar to that of Simon et al. (2016), estimating separate DiD regressions for financial outcomes and using SUR to formally test whether coverage gains jointly influence these financial indicators. By shifting from Simon et al.'s application of SUR on health behaviors to financial outcomes, we provide new evidence on whether coverage expansions under the ACA produced coordinated improvements across multiple dimensions of financial protection.

Section Three: Economic Theories and Hypothesis

This study's analysis of Medicaid expansion under the ACA is based on Arrow's (1963) market failure theory and Grossman's (1972) health capital model to collectively justify the policy's design and intervention. Arrow's (1963) seminal work identifies systemic inefficiencies that exist in healthcare markets: moral hazard (overconsumption due to insurance), adverse selection (high-risk individuals disproportionately seeking coverage), and information asymmetry (patients relying on providers for care decisions). Medicaid expansion directly addresses these issues by reducing financial barriers for low-income adults, correcting distorted pricings, and expanding access to care. Arrow's theory thus provides the *theoretical imperative* for government intervention and justifies Medicaid expansion as a necessary corrective mechanism to mitigate market inefficiencies.

Building on this foundation, Michael Grossman's (1972) health capital model provides the mechanism through which Medicaid expansion operates. By modeling health as a durable stock that depreciates over time but can be maintained through investments in medical care, Grossman's model demonstrates how insurance coverage influences health production decisions.

$$\pi = \frac{P_M}{\partial I/\partial T} + \frac{w}{\partial I/\partial M}$$

Where:

- π is the marginal cost of health investments.
- Pm is the price of medical care (e.g., OOP cost and premium); w is the wage rate.
- $\partial I/\partial T$ is the arginal productivity of time in health production.
- $\partial I/\partial M$ is the marginal productivity of medical care in improving health.

Subsidy on medical aid through ACA significantly reduces individual costs by slashing out-of-pocket (OOP) expenses (H3) and results in a net decrease in medical care price P_M for enrollees. This reduces the marginal cost of health investments π for individuals, which makes health investment more attractive and results in more people enrolling in the insurance by taking the advantage of cheaper health investments(H1). As enrollment increases, it dilutes risk in the health insurance pool, leading to a decrease in premiums (H2). Therefore, according to Grossman's framework, this should support our three hypotheses:

1) H1: ACA expansion increases insurance coverage.

2) H2: ACA expansion reduces premiums.

3) H3: ACA expansion reduces out-of-pocket costs.

Although the Grossman model provides valuable insights into health behavior, it makes several key assumptions: health is treated as a capital stock that depreciates at an exogenous, age-dependent rate, and individuals face no credit constraints or market imperfections. These assumptions introduce limitations. For instance, the model overlooks behavioral factors like bounded rationality, fails to account for uncertainty in health shocks or returns on health investments, and ignores institutional features. It also abstracts from social determinants of health and assumes an oversimplified, well-understood relationship between inputs (e.g., medical care) and health outcomes. Despite these limitations, Grossman's health investment model remains a seminal framework in health economics. It highlights the role of health in improving utility and productivity and provides a foundation for understanding why individuals allocate resources to health even in the absence of illness.

Gruber (2008) challenges the Medicaid expansion's impacts though an alternative theoretical consideration that public insurance expansions might crowd out private coverage. Gruber's analysis suggests that public insurance expansions may reduce private coverage through two mechanisms: (1) *direct substitution*, where individuals/families earning 100–138% of the federal poverty level (FPL) drop employer-sponsored insurance (ESI) to enroll in Medicaid, and (2) *employer response*, where firms reduce ESI offers knowing employees qualify for public options. However, Courtemanche et al. (2017) provide empirical evidence against significant crowd-out effects from Medicaid expansion. Their findings indicate that private insurance coverage *increased* alongside Medicaid enrollment in expansion states and suggest that expanded coverage lowers the effective price of health investments *without* reducing private insurance, which is consistent with Grossman's model.

Section Four: Methodology

Data Sources and Sample Selection

We draw on individual‑level microdata from the March ASEC of the CPS—a nationally representative survey widely used in labor and health economics conducted every March—and augment it with state‑level administrative data. Our analytic sample comprises non‑elderly adults aged 20–65 at the time of interview (which, after subtracting one year to align with the insurance‑coverage window, corresponds to ages 19–64). To assess pre‑trend validity and avoid potential instability associated with the early Trump administration, we pool CPS ASEC data from 2011 through 2017. Throughout, we apply the person‑level sampling weights (scaled to recover annual estimates) to preserve national representativeness.

We further restrict the sample by excluding individuals with missing values on key variables and drop proxy respondents and those in group quarters to ensure measurement consistency and comparability across states. After theis restrictions, our final sample includes approximately 786,258 observations across the states.

To account for macroeconomic variation across time and space—particularly differences in labor market recovery trajectories following the 2008–2009 financial crisis—we merge annual, seasonally adjusted state unemployment rates from the Bureau of Labor Statistics' Local Area Unemployment Statistics (LAUS). In addition, we incorporate information on state Medicaid expansion decisions and effective implementation dates from the Kaiser Family Foundation (KFF), cross-validated against timelines published in Courtemanche et al. (2017) and Simon (2016). These sources are widely recognized for their accuracy and reliability in health policy research.

While the CPS ASEC offers rich individual‑level data, it is still subject to some limitations. First, insurance coverage variables are self‑reported and refer to respondents' status one year prior to the survey, which can introduce recall error or misclassification—particularly for Medicaid, which is often underreported. To address this temporal mismatch, we subtract one year from both age and state of residence so that these variables align with the coverage window. Unfortunately, other covariates—marital status, household size, unemployment status, and educational attainment—cannot be similarly lagged, and we assume they remain unchanged over the recall period; we believe any resulting bias to be minimal. Second, the cross-sectional design of the CPS limits our ability to track individuals over time, although state-year variation supports a credible DiD framework. Third, ACA implementation timing may vary within states or overlap with other state-level policies not fully captured in our data; we mitigate this by including state and year fixed effects and by validating expansion dates using multiple sources. Fourth, small sample sizes in late-expanding states (e.g., Montana, Alaska) could reduce power to detect heterogeneous effects. Finally, although we cluster standard errors at the state level to account for intra-state correlation, this conservative approach may overstate uncertainty in contexts with limited state-level variation.

Model Specification

Our empirical strategy exploits a difference-in-differences (DiD) design, comparing changes in outcomes between expansion and non-expansion states before and after Medicaid expansion. We estimate all specifications using the CPS ASEC person weights, scaled by 1/7 (to reflect the seven years of pooled data and recover a per-year weight), and compute standard errors clustered at the state level to adjust for intra-state correlation. Our regressions include state and year fixed effects, the seasonally-adjusted state-level annual unemployment rate, and a rich set of individual covariates (age and state of residence lagged one year, gender, marital status, household size, race/ethnicity and education dummies, citizenship, student status, and continuous household income) to absorb time-invariant heterogeneity and observed confounders. We examine four outcome domains—insurance coverage, financial burden, premium payments, and health status—each estimated with the same survey-weighted linear regression (our "DiD" specification). Even when an outcome is binary (e.g., insured, zero-premium, MOOP ≥ 5 % income, good health), we use OLS (a linear probability model) to obtain unbiased estimates of the average treatment effect and to simplify interpretation.

Furthermore, to capture contemporaneous correlations across key outcomes, we estimate a Seemingly Unrelated Regression (SUR) system for the four primary endpoints—$Insured$, $MOOP \geq 5\,\%\,Income$, $Zero\,Premium$, and $Good\,Health$—using the identical right-hand-side specification. While we do not expect efficiency gains relative to separate DiD estimates (given the uniform covariates), the SUR framework allows us to test joint hypotheses on the cross-equation covariance matrix via an F-test and to better characterize any simultaneous policy effects.

In robustness checks, we validate the parallel‑trends assumption by estimating event‑study models that test for differential pre‑trends and by running placebo analyses over purely pre‑expansion periods. Although expansion timing was not randomized, prior work indicates that political considerations—rather than contemporaneous health or economic shocks—primarily drove adoption, which, once we condition on state and year fixed effects and our rich set of controls, mitigates concerns about endogenous policy timing. Because some states (for example, Alaska and Montana) expanded Medicaid later than others, this staggered timing creates additional 'before' and 'after' comparisons within those states—providing extra leverage to isolate the policy's effect

Section Five: Empirical Results

Descriptive statistics

Tables F1-F4 provide an overview of the sample characteristics, offering context on insurance coverage, financial outcomes and demographic composition, before and after Medicaid expansion across expansion and non-expansion states.

In Table F1-F2, while many covariates show statistically significant differences between expansion and non-expansion states prior to 2014, the magnitude of these differences is generally small. This suggests that selection into treatment is not random, but observable imbalance is limited, supporting the use of covariate adjustment and fixed effects in our regression framework.

Table F3 presents expansion-states exhibited higher insured rates even before the policy, and saw larger post-2014 increases in coverage, driven primarily by gains in Medicaid enrollment. Private insurance rates declined modestly across all groups, suggesting limited crowd-out effects. The descriptive patterns are consistent with expectations from ACA implementation.

Table F4 summarizes financial and health outcomes. Individuals in expansion states had slightly lower out-of-pocket medical expenses and a higher prevalence of zero-premium insurance plans, both before and after expansion. These differences widened post-2014, implying improved financial protection. Self-reported health measures were marginally better in expansion states, though differences were modest in size.

These descriptive findings provide preliminary evidence that Medicaid expansion was associated with improved coverage and affordability, though causal claims are addressed through regression analysis in the following sections.

DiD regression result

The results from the DiD regression, reported in Table F5, indicate that the Medicaid expansion significantly increased insurance coverage, consistent with the policy's intended impact. In the full sample, the expansion is associated with a 0.96 percentage point (pp) increase in the probability of being insured ($p < 0.01$). While statistically significant, this aggregate effect is economically modest. However, effects are substantially larger among the mid-FPL (3.02 pp) and low-FPL (4.29 pp) groups—populations directly targeted by the policy. These findings suggest that the average treatment effect masks important heterogeneity, and that the expansion's impact was more pronounced among lower-income individuals.

The gains in insurance coverage are driven primarily by increased Medicaid enrollment, which rose by 3.61 pp overall, while private insurance coverage fell by 1.67 pp. This substitution effect is particularly evident in the mid-FPL group, where Medicaid coverage rose by 9.03% and private coverage declined by 4.86%. Other forms of coverage remained unchanged. These results suggest that the ACA's design shifted uninsured individuals into public programs with limited disruption to private insurance markets.

<u>Financial Outcomes and Affordability</u>

Table F6 presents effects on financial protection. In the full sample, out-of-pocket (OOP) medical spending declined by $239, and the probability of high medical burden—defined as exceeding 5% or 10% of income—fell by 2.8 pp and 2.5 pp, respectively. These improvements are statistically significant and economically meaningful, especially among the mid-FPL group, where reductions are most pronounced. The disproportionate benefits to mid-FPL households likely reflect eligibility cliffs in the ACA subsidy structure, which limited cost-sharing benefits for individuals just above Medicaid thresholds.

In addition, the likelihood of enrolling in a zero-premium plan increased by 8.4 pp among mid-FPL individuals. This result supports the notion that coverage affordability improved significantly, enhancing access and reducing the financial risk of medical care.

<u>Health Outcomes</u>

Despite improvements in coverage and financial protection, the expansion is associated with a slight decline in self-reported health status across all groups. These changes, while statistically significant, are small in magnitude and may reflect increased diagnosis rates or shifts in subjective health perception, rather than true health deterioration. This interpretation is consistent with Grossman's (1972) health capital model, which posits that expanded access to care may lead to greater awareness of health conditions, thereby influencing reported health status.

Seemingly Unrelated Regression (SUR) Analysis

To account for possible cross-equation correlation among key outcomes, we estimate a weighted Seemingly Unrelated Regression (SUR) model. As shown in Table F7, the panel F-tests for residual independence are highly significant ($p < 0.001$) across all samples (full, mid-FPL, low-FPL), which confirms that residuals are correlated.

The significant cross-equation dependencies suggest that unobserved local factors—such as provider availability or state implementation quality—jointly influenced outcomes. These results strengthen the conclusion that Medicaid expansion had broad system-wide effects, rather than isolated impacts on specific outcomes.

Our findings are consistent with previous research. For instance, Courtemanche et al. (2019) similarly report that Medicaid expansion increased coverage significantly, particularly among low- and mid-FPL groups (3.02–4.29 pp gains), and induced a shift from private to public insurance—consistent with the ACA's income-targeted design. However, the modest aggregate effect (0.96 pp) is smaller than estimates for early expansion states, suggesting diminishing marginal returns as later-adopting states faced challenges in reaching harder-to-enroll populations.

The financial protection improvements—including the $239 reduction in OOP spending and 2.5–2.8 pp drop in high medical burden—are consistent with findings from Simon et al. (2016), but we observe heterogeneity that prior studies often averaged out. Mid-FPL households benefited more, likely due to the interplay between eligibility thresholds and subsidy tapering. The slight decline in self-reported health mirrors Grossman's theory that greater access leads to increased health knowledge and altered reporting, rather than actual health deterioration.

Finally, while the DiD design mitigates concerns about confounding from time-invariant unobserved differences, we acknowledge that concurrent state-level policies, economic shifts, or variation in policy implementation could bias estimates. The SUR results provide further justification for considering interdependencies across domains.

Robustness test

We estimate event study models to visualize dynamic treatment effects and assess the plausibility of the parallel trends assumption. Coefficients for the pre-treatment periods ($t = -3, -2$) are jointly tested. The results are presented in Table F8-10 and Figure F1.

For the full sample, outcomes show no significant pre-trend for insurance coverage, zero premium, or OOP burden, supporting the identifying assumption. Post-expansion, we observe a significant increase in self-reported good health, at $t = 1, 2, 3$. In the mid-FPL subgroup, we again find flat pre-trends. Medicaid expansion leads to substantial gains in zero-premium coverage, starting at $t = 0$ (which is 5.9%) and growing to 10.1% by $t = 3$. But there are no significant changes detected in insurance or health outcomes. In the low-FPL subgroup, insurance coverage increases significantly from $t = 0$, peaking at 5.4% at $t = 1$. Zero-premium coverage also rises significantly at $t = 1, 2, 3$(5.8%, 5.7% and 4.6% respectively). However, no statistically significant changes are observed for OOP $\geq$ 5% or good health, suggesting more limited effects for this group beyond insurance uptake.

Overall, the event study confirms the parallel trends assumption and highlights important heterogeneity in treatment effects: mid-FPL individuals experience the most consistent gains in affordability, while low-FPL individuals primarily benefit from increased coverage.

Section Six: Conclusion

Discussion

This study provides evidence that the ACA Medicaid expansions achieved their dual aims of increasing insurance coverage and financial protection for low-income populations.

Coverage Gains with Market Reallocation

Medicaid expansion led to a 3.02–4.29 percentage point (pp) increase in insurance coverage among low- and mid-FPL populations, primarily through growth in public insurance enrollment (+3.61 pp), accompanied by a modest decline in private coverage (−1.67 pp). This pattern is more consistent with targeted uptake by the previously uninsured than with crowd-out of existing private plans, aligning with the ACA's design to fill safety net gaps without undermining existing coverage sources. Notably, the 8.4 pp increase in zero-premium plan enrollment for the mid-FPL group emphasizes the importance of affordability—not just eligibility—in driving take-up. This finding supports behavioral economics research on inertia and the power of default options, such as that by Bhargava et al. (2017), and suggests that reducing or eliminating up-front costs is essential for maximizing enrollment.

Financial Protection with Heterogeneous Benefit

The policy also delivered measurable financial benefits, with reductions in average out-of-pocket (OOP) medical spending (−\$239) and the likelihood of facing high medical cost burdens (−2.5–2.8 pp). However, these gains were not evenly distributed. Mid-FPL households benefited more than their lower-income counterparts, likely due to sharp eligibility cliffs in the ACA's subsidy structure. These cliffs reduce the marginal value of additional support for the lowest-income individuals while making those just above the threshold more sensitive to cost-sharing burdens. These results build on the findings of Simon et al. (2016), providing new evidence on income-graded effects and reinforcing the broader challenge of designing equitable, means-tested policy interventions.

Health Outcomes and Moral Hazard

Although the expansion was associated with modest declines in self-reported health status, this counterintuitive result is consistent with previous literature (e.g., Kolstad & Kowalski, 2016) and likely reflects increased diagnostic activity rather than true health deterioration. As newly insured individuals gain access to care, they may receive diagnoses previously unrecognized or underreported, in line with Grossman's (1972) model of health capital, which predicts increased utilization in response to reduced care costs. In addition, insured respondents may recalibrate their self-assessments of health in light of professional evaluations, introducing reporting bias. These dynamics suggest that the health benefits of Medicaid expansion may be delayed and more observable in long-term preventive care outcomes.

The Seemingly Unrelated Regression (SUR) framework employed in this study captures the interdependence of outcomes across insurance, financial, and health domains. Joint significance tests (F-tests, $p < 0.001$) indicate that unobserved state-level factors—such as provider network capacity, administrative efficiency, and local healthcare infrastructure—simultaneously influence all three domains. This suggests the importance of modeling policy effects in a holistic, multivariate framework rather than in isolation.

Policy Implications

The findings of this study offer strong empirical support for the continued expansion of Medicaid as a targeted, effective policy to improve insurance coverage and financial protection among low-income populations. The substantial increases in public insurance enrollment—particularly among mid- and low-FPL groups—and the associated reductions in out-of-pocket medical spending and high-burden rates highlight Medicaid's capacity to alleviate financial stress for vulnerable households. Importantly, the observed rise in zero-premium plan take-up underscores the critical role of minimizing up-front costs to boost coverage rates. Policymakers in non-expansion states should consider these results when evaluating the costs and benefits of expanding Medicaid, particularly as evidence indicates that previously uninsured populations near eligibility thresholds stand to benefit the most. Moreover, the modest but consistent substitution from private to public insurance—without adverse spillovers to other forms of coverage—suggests that Medicaid expansion can proceed with minimal disruption to existing insurance markets.

From an implementation perspective, the heterogeneity in treatment effects by income group and the evidence of correlated outcome shifts across insurance, financial, and health domains highlight the importance of integrated policy design. Practical efforts to maximize the benefits of expansion should focus on outreach to eligible but unenrolled populations, tailoring enrollment strategies to account for local administrative capacity and healthcare infrastructure. For example, enhancing access in rural or provider-shortage areas may require complementary investments in telehealth services or provider reimbursement incentives. In addition, the potential for increased diagnosis rates among new enrollees suggests a need for strengthening preventive care systems and chronic disease management programs to ensure that expanded access translates into sustained health improvements.

Future Research Directions

This study highlights several critical directions for future research on Medicaid expansion. First, methodological improvements—such as synthetic control methods or covariate balancing—could strengthen causal inference by addressing pre-treatment imbalances in financial outcomes (e.g., out-of-pocket spending, zero-premium enrollment). Second, local heterogeneity in policy implementation warrants deeper investigation. Our state-level analysis obscures meaningful variation in rural/urban rollout, provider networks, and regional healthcare infrastructure; county- or hospital-referral-region-level studies could identify context-specific drivers of success. Third, health outcome measures should be refined. Linking survey data to Medicaid claims or clinical registries would mitigate biases in self-reported health status and enable analysis of preventive care utilization, chronic disease management, and objective clinical outcomes. Fourth, alternative identification strategies (e.g., instrumental variables exploiting exogenous funding variations) could disentangle Medicaid expansion's effects from concurrent state policies or economic shocks. Finally, the long-term sustainability of coverage gains, financial protections, and health impacts—particularly in late-adopting states—remains an open question requiring extended longitudinal analysis. Together, these directions would provide a more nuanced understanding of the policy's effectiveness across diverse populations and settings.

Limitations

While this study provides robust evidence on the effects of Medicaid expansion, several limitations warrant consideration. First, although the difference-in-differences design accounts for secular trends, pre-treatment imbalances in zero-premium enrollment and out-of-pocket spending raise concerns about the validity of the parallel trends assumption. While event study results largely support the design, some residual bias may persist in estimating financial outcomes. Future research could apply alternative methods—such as synthetic control or covariate balancing—to address this concern more rigorously. Second, the use of state-level variation may obscure important within-state heterogeneity in policy implementation. Differences in urban versus rural provider networks, county-level administrative capacity, and regional healthcare market structures suggest that finer-grained analyses at the county or hospital referral region level could yield additional insight. Third, the reliance on CPS data limits our health outcome measurement to self-reported status, without information on utilization or clinical indicators, making it difficult to distinguish true health improvements from increased diagnosis, assess preventive care uptake, or evaluate chronic disease management. Linking survey data to Medicaid claims or clinical registries would enable richer health outcome analysis in future work. Fourth, despite employing Seemingly Unrelated Regression and extensive controls, we cannot fully rule out confounding from concurrent state policy changes (e.g., reinsurance programs), local labor market shocks, or provider-side dynamics such as hospital consolidation. Future studies might benefit from instrumental variable strategies leveraging exogenous variation in federal Medicaid matching rates. Fifth, our sample excludes several vulnerable populations—including undocumented immigrants, marginally housed individuals, and incarcerated populations—whose healthcare needs are substantial but underrepresented in survey data. These exclusions limit the generalizability of our findings to the most disadvantaged groups. Finally, our study period (2011–2018) captures only medium-term effects of expansion; the long-term durability of coverage gains and health improvements—especially in late-adopting states—remains an important direction for future research.

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Appendix

Appendix A: Treatment Variable Construction

Following established literature (e.g., Courtemanche et al., 2017; Simon, 2016), we define a binary treatment variable, $Treatment_s$, which equals 1 if the states had implemented the ACA Medicaid expansion before 2017, and 0 otherwise. It is noteworthy that some states that adopted the expansion after the federal 2014 baseline are late adopters, with implementation dates as follows: Michigan (April 2014), New Hampshire (August 2014), Pennsylvania (January 2015), Indiana (February 2015), Alaska (September 2015), Montana (January 2016), and Louisiana (July 2016). The following Table A1 is the full list of expansion states ($Treatment_s$=1)and non-expansion states($Treatment_s$=0). We also define a binary treatment variable, $Post_t$, which equals 1 if the state had implemented the ACA Medicaid expansion as of survey year t, regardless of the month that took effect, and 0 otherwise.

Table A1: Classification of Expansion and Non-Expansion States

Expansion States (2014–2018)	**Non-Expansion States**
Arizona	Alabama
Arkansas	Florida
California	Georgia
Colorado	Idaho
Connecticut	Kansas
Delaware	Maine
District of Columbia	Mississippi
Hawaii	Missouri
Illinois	Nebraska
Iowa	North Carolina
Kentucky	Oklahoma
Maryland	South Carolina
Massachusetts	South Dakota
Minnesota	Tennessee
Nevada	Texas
New Jersey	Utah
New Mexico	Virginia
New York	Wisconsin
North Dakota	Wyoming
Ohio	
Oregon	
Rhode Island	
Vermont	
Washington	
West Virginia	
Michigan (Apr 2014)	
New Hampshire (Aug 2014)	
Pennsylvania (Jan 2015)	
Indiana (Feb 2015)	
Alaska (Sep 2015)	
Montana (Jan 2016)	
Louisiana (Jul 2016)	

Notes: This table shows the state classification as regards Medicaid eligibility for adults, including 50 states as well as the District of Columbia.

Appendix B: Outcome Variable Construction

We examine four main categories of outcomes:

1. **Insurance coverage**: measured through binary indicators for any coverage ($Insured$), private coverage ($Private\ Insurance$), which is defined as either employment sponsored or directly purchased insurance, Medicaid enrollment ($Medicaid$), and other coverage ($Other\ Coverage$), which is defined as neither covered by private insurance nor covered by Medicaid. It is note-worthy that the latter three are not mutually exclusive.

2. **Financial burden**: proxied by total out-of-pocket medical spending, which is the total amount the family actually paid for medical services ($MOOP\ Expense$) and two binary indicators for whether MOOP exceeds 5% ($MOOP \geq 5\%\ Income$) and 10% ($MOOP \geq 5\%\ Income$) of household income. The 5% threshold is used to reflect affordability standards in the literature (Baird, 2016).

3. **Premium payments**: measured using family-level premium payments ($HIPVAL$) and a binary indicator for zero-premium coverage ($Zero\ Premium$), suggested by (Abramowitz, 2020).

4. **Health status**: based on self-assessed health, both as a five-point ordinal variable and as a binary indicator ($Good\ Health$) equal to 1 if the respondent reports "good," "very good," or "excellent" health.

Appendix C: Subgroup construction

As a measure of socioeconomic status, we calculate each household's income-to-poverty ratio based on the Department of Health and Human Services' Federal Poverty Guidelines, which adjust for household size and year. For stratified analyses, we use the following standard poverty thresholds:

- Low-FPL (< 100%):
  - In expansion states, individuals become eligible for Medicaid.
  - In non-expansion states, individuals remain ineligible for both Medicaid and subsidized exchange coverage (the "coverage gap").
- Mid-FPL (100 – 138%):
  - In expansion states, individuals gain Medicaid coverage.
  - In non-expansion states, individuals qualify for subsidized exchange plans (e.g., premium tax credits on ACA marketplaces).

We also treat the income-to-poverty ratio as a continuous covariate in our regressions to capture fine-grained variation. Ultimately, we define three analysis subgroups:

1. $Full\ sample$ (all income levels)
2. $Mid - FPL$ (100–138%)
3. $Low - FPL$(below 100%)

Appendix D: Control Variables construction

We control for a rich set of individual‑level characteristics: age and state of residence (both lagged by one year to align with insurance questions about "one year ago"), gender, marital status, household size, and current unemployment status. We include race/ethnicity indicators for Hispanic, non-Hispanic Black, Asian, and Other (with non-Hispanic White as the omitted category), and educational attainment dummies for high-school graduate, some college, and college graduate (below high school is the reference group). Continuous household income is also included. To capture local labor-market conditions, we add the seasonally adjusted, state-level annual unemployment rate. Finally, state and year fixed effects absorb time-invariant regional heterogeneity and common national shocks.

Because insurance coverage is reported with a one-year lag, we adjust both age and state of residence accordingly; unfortunately we cannot similarly lag marital status, household size, unemployment status, or educational attainment, so we assume these remain unchanged over the recall period and believe any resulting bias to be minimal. These covariates follow the specifications in Courtemanche et al. (2017) and Simon et al. (2016).

Appendix E: Empirical Strategy

DiD framework

We estimate the causal effects of Medicaid expansion using a two-way fixed-effects difference-in-differences (DiD) framework. We estimate survey-weighted linear (probability) regression models specified as:

$$Y_{ist} = \alpha + \beta\,(Treatment_s \times Post_t) + \gamma X'_{ist} + \eta UnempRate_{st} + \delta State_s + \theta Year_t + \varepsilon_{ist}$$

Where:

- Y is the outcome for individual i in state s and year t,
- $Treatment \times Post$ is the variable of interest,
- X'_{ist} is the control vector, δ_s are state fixed effects, τ_t are year fixed effects, and ε_{ist} is the error term.
- Outcomes are modeled with weighted OLS on the same right-hand side variables. Standard errors are clustered at the state level to account for within-state correlation.

Economic Mechanism & Joint Outcomes

Grossman's health-capital framework implies coverage, premiums, and out-of-pocket (OOP) costs are jointly determined. Subsidizing coverage lowers OOP costs, boosts health-capital investments, enlarges the risk pool (reducing adverse selection and premiums), and further incentivizes enrollment. Because unobserved factors (e.g., provider density) affect all outcomes, we estimate a system of Seemingly Unrelated Regressions (SUR) and perform a Breusch–Pagan test for residual covariance—rejecting independence among all outcomes.

Event-Study Specification.

To visualize dynamics and test pre-trends, we estimate as the following:

Let τ be the calendar year in which the state s first expands Medicaid. Define the relative-timing indicators

$$D_{s,t+k} = 1\{t = \tau_s + k\}\ (k = -K, \ldots, +K)$$

and omit $k = -1$ as the reference period.

Then our event-study DiD reads:

$$Y_{ist} = \alpha + \sum_{k=-K,\, k\neq -1}^{K} \beta(Treatment_s \times D_{s,\,t+k}) + X'_{ist}\gamma + \delta_s + \tau_t + \varepsilon_{ist}$$

Where:

- $Treatment_s$ is a time-invariant 0/1 indicator for ever-expansion.
- $D_{s,\,t+k}$ picks out the year k relative to that state's expansion.
- The coefficients $\beta_{k<0}$ test for any pre-trends, while $\beta_{k>0}$ trace out the dynamic post-expansion effects.
- State fixed effects $D_{s,\,t+k}$ eliminate time-invariant confounders; year fixed effects $D_{s,\,t+k}$ absorb common shocks. X'_{ist} is the control vector, δ_s are state fixed effects, τ_t are year fixed effects, and ε_{ist} is the error term.
- Outcomes are modeled with weighted OLS on the same right-hand side variables. Standard errors are clustered at the state level to account for within-state correlation.

Identification & Robustness

- Selection bias & heterogeneity: Expansion may correlate with unobservables that also affect outcomes. We mitigate this by controlling for rich observables and fixed effects, conducting subgroup analyses (e.g., by income or state economic conditions), and running event study tests in pre-expansion years.
- Parallel trends: We implement an event-study specification to test for differential pre-trends (see below).
- Endogeneity: Policy adoption could be endogenous if states expanded Medicaid in response to unobserved shocks—say, sudden increases in illness or financial distress—thereby biasing our estimates. To address this, we leverage staggered rollout timing across states and argue that political timing (e.g., legislative calendars, gubernatorial priorities) drove expansion dates more than short-run health or economic shocks.
- Simultaneity/reverse causality: If worsening outcomes spur expansion, DiD may overstate effects. Fixed effects absorb time-invariant heterogeneity; year effects absorb common shocks; and our event-study tests look for anticipatory changes in outcomes that would signal reverse causality.

Appendix F: Table and figures

Tables

Table F1 Descriptive statistics for demographic variables

Panel 1: Demographics	Expansion states		Non-expansion states		Pre-2014
	2011–13	2014–17	2011–13	2014–17	difference
Age	41.18	41.24	41.03	41.22	0.15***
	(12.68)	(12.80)	(12.62)	(12.75)	
Household Income	91,116.16	103,055.95	80,387.40	91,194.79	10,728.76***
	(94,738.60)	(106,933.94)	(83,349.69)	(97,525.79)	
Female	0.52	0.52	0.52	0.52	0.00
	(0.50)	(0.50)	(0.50)	(0.50)	
Married	0.57	0.57	0.60	0.59	0.03***
	(0.49)	(0.50)	(0.49)	(0.49)	
Family Size	3.18	3.18	3.14	3.15	0.04***
	(1.63)	(1.64)	(1.58)	(1.58)	
Unemployed	0.28	0.27	0.28	0.27	0.01***
	(0.45)	(0.44)	(0.45)	(0.44)	

Notes: Each cell reports mean (standard deviation), unweighted. The Pre-2014 difference is the mean difference between expansion and non-expansion states in 201113. Stars indicate significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table F2 Descriptive statistics for demographic variables (Cont.)

Panel 1: Demographics (continued)	Expansion states		Non-expansion states		Pre-2014
	2011–13	2014–17	2011–13	2014–17	difference
Race					
Hispanic	0.17	0.20	0.17	0.19	0.00
	(0.38)	(0.40)	(0.38)	(0.39)	
Black	0.10	0.10	0.14	0.15	0.04***
	(0.30)	(0.30)	(0.35)	(0.36)	
Asian	0.07	0.08	0.03	0.03	0.04***
	(0.26)	(0.28)	(0.17)	(0.18)	
Other Race	0.03	0.04	0.03	0.03	0.01***
	(0.18)	(0.19)	(0.17)	(0.16)	
Education / Citizenship					
HS Grad	0.28	0.28	0.29	0.29	0.01***
	(0.45)	(0.45)	(0.46)	(0.45)	
Some College	0.29	0.29	0.31	0.31	0.02***
	(0.46)	(0.45)	(0.46)	(0.46)	
College Grad	0.32	0.34	0.28	0.30	0.04***
	(0.47)	(0.47)	(0.45)	(0.46)	
Foreign Born	0.20	0.22	0.16	0.17	0.04***
	(0.40)	(0.41)	(0.37)	(0.38)	
U.S. Citizen	0.90	0.89	0.91	0.91	0.01***
	(0.30)	(0.31)	(0.29)	(0.29)	
Student	0.07	0.07	0.06	0.07	0.01***
	(0.25)	(0.26)	(0.24)	(0.25)	

Notes: Each cell reports mean (standard deviation), unweighted. The Pre-2014 difference is the mean difference between expansion and non-expansion states in 201113. Stars indicate significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table F3 Descriptive statistics for insurance variables

Panel 2: Insurance	**Expansion states** 2011–13	2014–17	**Non-expansion states** 2011–13	2014–17	**Pre-2014** difference
Insured	0.83	0.90	0.78	0.84	0.05***
	(0.38)	(0.30)	(0.41)	(0.36)	
Private Insurance	0.70	0.73	0.67	0.71	0.03***
	(0.46)	(0.44)	(0.47)	(0.45)	
Medicaid	0.12	0.18	0.09	0.10	0.04***
	(0.33)	(0.38)	(0.28)	(0.30)	
Other Coverage	0.04	0.04	0.05	0.06	-0.01***
	(0.19)	(0.20)	(0.22)	(0.23)	

Notes: Each cell reports mean (standard deviation), unweighted. The Pre-2014 difference is the mean difference between expansion and non-expansion states in 2011–13. Stars indicate significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F4 Descriptive statistics for MOOP, premium and health variables

Panel 3: Health, MOOP, Premium	**Expansion states** 2011–13	2014–17	**Non-expansion states** 2011–13	2014–17	**Pre-2014** difference
MOOP Expense ($)	4,003.66	4,634.42	4,118.23	4,839.87	-114.57***
	(6,261.00)	(8,196.42)	(7,075.98)	(8,497.16)	
MOOP $\geq$ 5% Income	0.36	0.37	0.41	0.42	-0.05***
	(0.48)	(0.48)	(0.49)	(0.49)	
MOOP $\geq$ 10% Income	0.17	0.18	0.21	0.22	-0.03***
	(0.38)	(0.38)	(0.40)	(0.41)	
HIPVAL ($)	2,098.54	2,664.70	2,062.51	2,657.04	36.03***
	(3,744.53)	(5,415.41)	(3,339.20)	(4,388.82)	
Zero Premium	0.43	0.36	0.42	0.33	0.01***
	(0.50)	(0.48)	(0.49)	(0.47)	
Health Score	3.78	3.78	3.73	3.76	0.05***
	(1.05)	(1.04)	(1.07)	(1.05)	
Good Health	0.89	0.89	0.88	0.88	0.01***
	(0.32)	(0.31)	(0.33)	(0.32)	

Notes: Each cell reports mean (standard deviation), unweighted. The Pre-2014 difference is the mean difference between expansion and non-expansion states in 2011–13. Stars indicate significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F5 Survey weighted linear regression results for insurance variables

Panel 1: Insurance	Full sample	Mid-FPL	Low-FPL
Insured	0.010	0.030*	0.043***
	(0.007)	(0.018)	(0.015)
Private Insurance	–0.017***	–0.049***	–0.015
	(0.006)	(0.013)	(0.009)
Medicaid	0.036***	0.090***	0.066***
	(0.005)	(0.012)	(0.013)
Other Public Coverage	0.000	0.004	–0.003
	(0.002)	(0.007)	(0.005)

Notes: Coefficients and standard errors of survey-weighted linear probability models, with estimation of state-level variance. Controls include individual covariates, state and year fixed effects, and state unemployment rate, all weighted by ASECWT_adj. Full sample $N = 786,258$; Mid-FPL $N = 49,275$; Low-FPL $N = 106,096$. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table F6 Survey weighted linear regression results for MOOP, premium and health variables

Panel 2: MOOP, Premium, and Health	Full sample	Mid-FPL	Low-FPL
Medical Out-of-Pocket			
OOP Expense ($)	86.043	–239.179**	–3.952
	(73.457)	(101.461)	(65.456)
MOOP ≥ 5% Income	0.007*	–0.028*	–0.011
	(0.004)	(0.015)	(0.011)
MOOP ≥ 10% Income	0.001	–0.025**	–0.008
	(0.004)	(0.011)	(0.010)
Health Insurance Premium			
Premium Value ($)	66.748	–178.349**	–9.436
	(65.632)	(80.268)	(39.075)
Zero Premium	0.014**	0.084***	0.041***
	(0.007)	(0.014)	(0.014)
Health Status			
Health Score	–0.048***	–0.062*	–0.065**
	(0.009)	(0.031)	(0.025)
Good Health	–0.007**	–0.012	–0.011
	(0.003)	(0.008)	(0.009)

Notes: Coefficients and standard errors of survey-weighted linear probability models, with estimation of state-level variance. Controls include individual covariates, state and year fixed effects, and state unemployment rate, all weighted by ASECWT_adj. Full sample $N = 786,258$; Mid-FPL $N = 49,275$; Low-FPL $N = 106,096$. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table F7 Seemingly unrelated regression results

	Full-Sample		Mid-FPL		Low-FPL	
	F-stat	p-value	F-stat	p-value	F-stat	p-value
Economic outcome	134960.48	$<$0.001	13721.31	$<$0.001	12652.09	$<$0.001
Full outcome	145628.79	$<$0.001	15488.12	$<$0.001	13820.28	$<$0.001

Note: F-tests for zero residual covariances of survey-weighted Seemingly Unrelated regression, with estimation of state-level variance. Controls include individual covariates, state and year fixed effects, and state unemployment rate, all weighted by ASECWT_adj. Full sample $N = 786,258$; Mid-FPL $N = 49,275$; Low-FPL $N = 106,096$.

Table F8 Event study results for Full Sample

Full Sample	t = -3	t = -2	t = 0	t = 1	t = 2	t = 3	p-value
Insured	-0.013	-0.004	0.005	0.009	0.001	0.007	0.238
	(0.007)	(0.005)	(0.006)	(0.008)	(0.009)	(0.009)	
OOP $\geq$ 5% Income	0.001	0.003	0.013*	-0.001	0.004	0.018	0.917
	(0.010)	(0.009)	(0.006)	(0.007)	(0.007)	(0.010)	
Zero Premium	0.010	-0.007	0.009	0.026*	0.015	0.010	0.108
	(0.009)	(0.008)	(0.006)	(0.010)	(0.009)	(0.010)	
Good Health	0.004	0.003	0.021	0.058**	0.057**	0.046*	0.072
	(0.011)	(0.009)	(0.015)	(0.016)	(0.020)	(0.020)	

Notes: Coefficients and standard errors of survey-weighted event study models, with estimation of state-level variance. The p-value column reports the test of the joint null hypothesis that all pre-expansion interaction coefficients (i.e. those with $t < 0$) are equal to zero. Controls include individual covariates, state and year fixed effects, and state unemployment rate, all weighted by ASECWT_adj. Full sample $N = 786,258$. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table F9 Event study results for Mid-FPL Subgroups

Mid-FPL Subgroup	t = -3	t = -2	t = 0	t = 1	t = 2	t = 3	p-value
Insured	-0.011	-0.002	0.022	0.031	0.031	0.026	0.819
	(0.019)	(0.018)	(0.021)	(0.022)	(0.024)	(0.028)	
OOP $\geq$ 5% Income	0.021	0.004	-0.011	-0.043	-0.025	-0.010	0.429
	(0.020)	(0.025)	(0.023)	(0.026)	(0.029)	(0.023)	
Zero Premium	0.001	-0.005	0.059^{**}	0.096^{**}	0.093^{**}	0.101^{***}	0.920
	(0.020)	(0.024)	(0.021)	(0.030)	(0.030)	(0.022)	
Good Health	0.000	-0.019	-0.019	-0.018	-0.012	-0.017	0.518
	(0.015)	(0.018)	(0.015)	(0.013)	(0.016)	(0.017)	

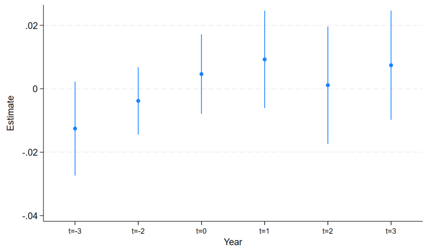
Notes: Coefficients and standard errors of survey-weighted event study models, with estimation of state-level variance. The p-value column reports the test of the joint null hypothesis that all pre-expansion interaction coefficients (i.e. those with $t < 0$) are equal to zero. Controls include individual covariates, state and year fixed effects, and state unemployment rate, all weighted by ASECWT_adj. Mid-FPL subgroup $N = 49,275$. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table F10 Event study results for Low-FPL Subgroups

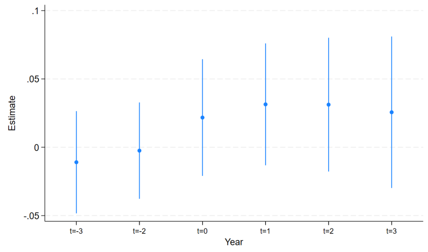
Low-FPL Subgroup	t = -3	t = -2	t = 0	t = 1	t = 2	t = 3	p-value
Insured	-0.002	-0.008	0.027^{*}	0.054^{**}	0.038	0.050^{*}	0.765
	(0.014)	(0.012)	(0.013)	(0.017)	(0.020)	(0.019)	
OOP $\geq$ 5% Income	0.010	0.008	-0.003	-0.015	-0.020	0.014	0.749
	(0.013)	(0.014)	(0.016)	(0.020)	(0.017)	(0.019)	
Zero Premium	0.004	0.003	0.021	0.058^{**}	0.057^{**}	0.046^{*}	0.923
	(0.011)	(0.009)	(0.015)	(0.016)	(0.020)	(0.020)	
Good Health	0.011	0.006	0.000	-0.020	-0.006	0.000	0.693
	(0.013)	(0.012)	(0.012)	(0.013)	(0.016)	(0.018)	

Notes: Coefficients and standard errors of survey-weighted event study models, with estimation of state-level variance. The p-value column reports the test of the joint null hypothesis that all pre-expansion interaction coefficients (i.e. those with $t < 0$) are equal to zero. Controls include individual covariates, state and year fixed effects, and state unemployment rate, all weighted by ASECWT_adj. Low-FPL subgroup $N = 106,096$. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

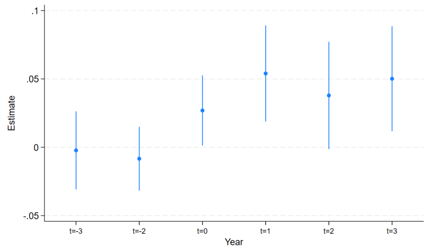
<u>Figures</u>



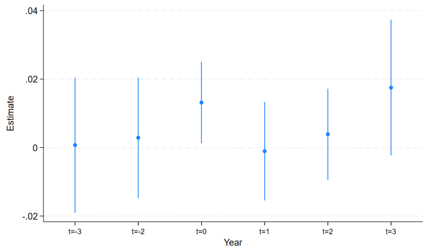
Insured (Full Samle)



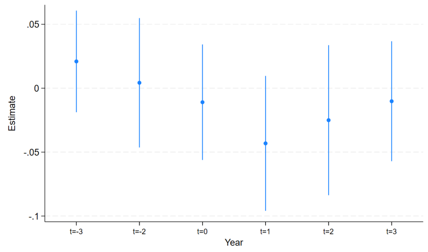
Insured (Mid-FPL)



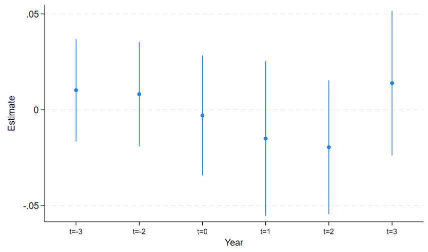
Insured (Low-FPL)



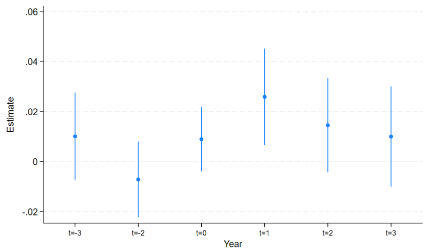
MOOP ≧ 5% Income (Full Sample)



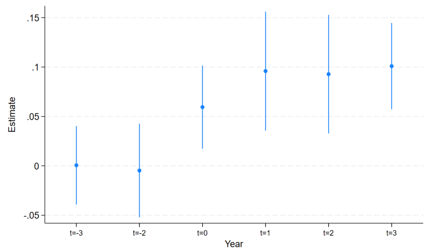
MOOP ≧ 5% Income (Mid-FPL)



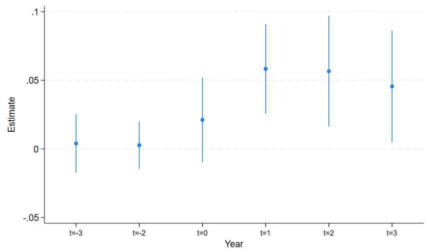
MOOP ≧ 5% Income (Low-FPL)



Zero HIPVAL (Full Sample)

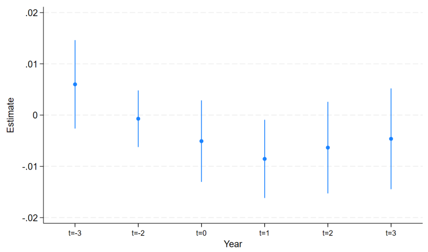


Zero HIPVAL (Mid-FPL)

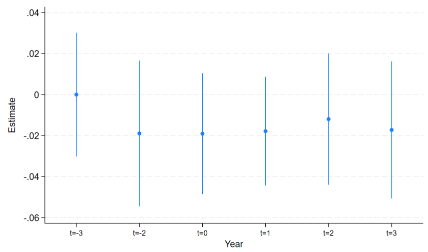


Zero HIPVAL (Low-FPL)

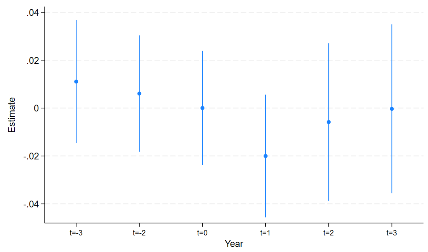
Goo



Health (Full Sample)



Good Health (Mid-FPL)



Good Health (Low-FPL)

Figure F1 Event study graphical results

Appendix G: Software and Reproducibility

Data processing and initial weighting checks are carried out in R 4.3 using the ipumsr, survey, and haven packages. All regression analyses—including LPM, SUR, and event-study models—are implemented in Stata 17.

Appendix H: Use of AI Assistance

This paper was drafted with structural and editorial assistance from OpenAI's ChatGPT. All substantive modeling decisions, data construction, and estimation commands were conceived and executed by the authors themselves.

Appendix I: Contribution

Table I1 Author Contributions

Role	Sing-Hao Ku	Pengzhen Wang	Wenqing Zhang
Conceptualization	✓	✓	
Software	✓		
Literature review	✓	✓	
Data curation	✓		
Formal Analysis	✓	✓	
Methodology	✓	✓	
Project Administration	✓	✓	✓
Resources	✓	✓	
Supervision	✓	✓	
Validation	✓	✓	
Visualization	✓	✓	
Writing - Original Draft	✓	✓	✓
Writing - Review & Editing	✓	✓	✓